

storage gives space to store documents on the Web.

Building a solution for the masses is not an easy task for any team. And the people at NeGD, too, faced many challenges while designing DigiLocker.

“Although the open source projects provide you ready solutions, you have to own them completely to be able to scale them for millions of users,” says Ranjan, while describing the prime challenge in deploying open source solutions for DigiLocker.

To solve the problem of scale, the NeGD team divided the entire project into certain phases. “The phases provide concrete goals regarding the number of users and files. Additionally, the infrastructure is constantly monitored, adjusted and fine-tuned with every major release. Scaling an application is an ongoing process, and needs constant changes in architecture and the solutions used. At the same time, we realise that every primary issuer on DigiLocker brings unique challenges, and we learn new lessons now and then,” Ranjan states.

Security measures to deliver a safe and secure platform

DigiLocker comes with Aadhaar integration to offer citizens a secured solution to store e-documents online. Also, there is the eSign option to let users self-attest their documents. The NeGD team uses some other security measures to make DigiLocker a safe and secure platform for the public.

“We are taking all the precautionary measures to ensure data is protected and uncompromised,” Ranjan asserts. The platform follows the OWASP (Open Web Application Security Project) security standards and guidelines. Additionally, there is a 256-bit SSL encryption layer on the server that is hosted in an ISO 27001 security-certified data centre. Data is regularly backed up with proper redundancy, and a one time password (OTP) is generated at the time of each sign-up to authenticate users. Security audits have also been conducted by a recognised audit agency to ensure safety and security.

“We follow standard software development practices of uniform coding standards, guidelines and reviews. Every product release is reviewed and tested internally for security vulnerabilities before it is deployed,” says Ranjan.

Open source technologies power this solution for the masses

DigiLocker is based on open source platforms including PHP, Python and Node.js. On the server front, there is Nginx and Apache, while MongoDB is used to enable the gateway access and MariaDB is deployed for user account related metadata.

“Open source technology gives you the freedom to try, test and scale your solution, one step at a time. One has so much choice of open source products and frameworks to choose from nowadays,” says Amit Jain, product manager.

The entire platform that brings the cloud storage solution to millions of Indian citizens is based on ownCloud Server. In fact, DigiLocker is supposedly the largest installation of ownCloud Server.



The way DigiLocker works

“With over two million users, DigiLocker is the largest installation of ownCloud Server worldwide. This demonstration of the capability of community software has created excitement in the ownCloud community. We are in constant touch with the ownCloud team and provide inputs for the community from time to time. We are also working on making DigiLocker source code available to the community under open source license,” says Amit Savant, technical product manager, NeGD.

Instead of proprietary solutions, the team led by Nayak opted for open source technologies to easily scale DigiLocker. “It would have been a mammoth task to deploy a proprietary solution of the scale of DigiLocker, which is meant for a billion people, if it succeeds,” Nayak says.

As the number of users grows, the NeGD team aims to scale up DigiLocker using more open source solutions. “Scaling up the application will be a constant endeavour with the growing number of issuers and requestors in the DigiLocker ecosystem. We are always experimenting with newer technologies that will help us make DigiLocker better,” Savant says.

The platform already has frameworks like Nginx and Memcached that offer high scalability. However, the team at NeGD’s New Delhi headquarters is planning to expand its existing coverage by deploying more community-based technologies.

Savant says there are plans to use only MongoDB for the gateway engine of the platform. “We have already developed an academic repository for DigiLocker, using MongoDB, and have developed the necessary skills within the team. We now think we are ready to take these skills to the next level,” he adds.

In addition to the open source NoSQL database for the gateway, the team is set to implement an OpenStack-based cloud for the dynamic computing environment. Deployment of GlusterFS—apart from RabbitMQ for messaging and GearMan for backend job processing—is in the pipeline.

“Open source technologies are better suited for government operations. These technologies can be customised and scaled in-house to suit your requirements. You are not locked in with one software vendor for the life of a project,” Nayak concludes. 

By: Jagmeet Singh

The author is an assistant editor at EFY.